

1. $C_i \in \{\Omega, \omega, i\}$ 이고, $C_k \in \{a.e., M_0\}$ 인 경우를 Note에 따라 Case 2와 같다.

$$\vec{x} = \vec{x}(x_1, x_2, x_3) \in \vec{x} \text{와 } \vec{x} \text{가 (a.e. } M(\nu)) \text{에 관한 항수인 것을 명두에 두고 플면}$$

Step 1) $\hat{x} = R_{[pqw]}^{-1} \hat{r}_{pqw} =$

$$\dot{\alpha} = R \begin{bmatrix} \dot{\alpha}_1 \\ \dot{\alpha}_2 \\ \dot{\alpha}_3 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \begin{bmatrix} \dot{\alpha}_1 \\ \dot{\alpha}_2 \\ \dot{\alpha}_3 \end{bmatrix}$$

Step 2) Lagrange Bracket $[c_i, c_k] =$

$$= \frac{\partial \tilde{x}}{\partial c_i} \frac{\partial \tilde{x}}{\partial c_k} - \frac{\partial \tilde{x}}{\partial c_k} \frac{\partial \tilde{x}}{\partial c_i} \quad \text{①} \quad \text{②} \quad \text{③} \quad \text{④}$$

$$\frac{\partial \bar{x}}{\partial c_i} + \frac{\partial \bar{x}}{\partial c_k}$$

$$\frac{\partial \vec{r}}{\partial C_i} = \left\{ \frac{\partial x_1}{\partial C_i}, \frac{\partial x_2}{\partial C_i}, \frac{\partial x_3}{\partial C_i} \right\}$$

$$\frac{\partial x_j}{\partial C_i} = \frac{\partial}{\partial C_i} \left(\frac{3}{I_{i=1}} R_{j,i} v_i \right) = \frac{3}{I_{i=1}} \left(\frac{\partial}{\partial C_i} R_{j,i} \right) \cdot v_i + \frac{3}{I_{i=1}} R_{j,i} \cdot \left(\frac{\partial}{\partial C_i} v_i \right) \quad \text{또 정리할 수 있다}$$

여기서 R 행렬은 (Ω, ω, j) 에 관한 함수로 $C_j \in \{\Omega, \omega, j\}$ 로 미분할 수 있지만,

$$\bar{r} = \bar{r}(a.e. M(\nu)) \text{ 이므로 } C_1 \text{로 미분할 수 있으므로 } C_1 \text{ 항은 소거된다}$$

$$\frac{\partial x_j}{\partial C_i} = \sum_{k=1}^3 \left(\frac{\partial R_{jk}}{\partial C_i} \right) \cdot r_k = \frac{\partial}{\partial C_i} R_{j1} \cdot r_1 + \frac{\partial}{\partial C_i} R_{j2} \cdot r_2 + \frac{\partial}{\partial C_i} R_{j3} \cdot r_3$$

$$= \frac{\partial \tilde{\mathcal{L}}}{\partial C_k} \left(\sum_{j=1}^3 R_{j\ell} V_\ell \right) = \sum_{j=1}^3 \left(\frac{\partial}{\partial C_k} R_{j\ell} \right) \cdot V_\ell + \sum_{j=1}^3 R_{j\ell} \left(\frac{\partial}{\partial C_k} V_\ell \right)$$

위와 마찬가지로 $R(\Sigma, \omega, i)$ 는 $c_k \in \{a, e, M_o\}$ 로 미판할 수 있다.

$$\vec{v} = \dot{\vec{x}}(a.e. M(v)) \text{ 만 양기되어 이번에는 양 항이 소거된다}$$

$$\frac{\partial \bar{x}}{\partial C_k} = \sum_{j=1}^3 R_{j,k} \left(\frac{\partial}{\partial C_k} V_j \right) + R_{j,3} \left(\frac{\partial}{\partial C_k} V_3 \right)$$

같은 방법으로 ③과 ④를 정리하면

$$\textcircled{3} \quad \frac{\partial \vec{r}}{\partial C_k} = \frac{\partial}{\partial C_k} \left(\sum_{i=1}^3 R_{ik} \vec{r}_i \right) = \sum_{i=1}^3 \left(\frac{\partial}{\partial C_k} R_{ik} \right) \cdot \vec{r}_i + \sum_{i=1}^3 R_{ik} \left(\frac{\partial}{\partial C_k} \vec{r}_i \right) = R_{j1} \left(\frac{\partial}{\partial C_k} \vec{r}_1 \right) + R_{j2} \left(\frac{\partial}{\partial C_k} \vec{r}_2 \right) + R_{j3} \left(\frac{\partial}{\partial C_k} \vec{r}_3 \right)$$

$$\textcircled{4} \quad \frac{\partial \vec{r}}{\partial C_i} = \frac{\partial}{\partial C_i} \left(\sum_{j=1}^3 R_{ij} \vec{r}_j \right) = \sum_{j=1}^3 \left(\frac{\partial}{\partial C_i} R_{ij} \right) \cdot \vec{r}_j + \sum_{j=1}^3 R_{ij} \left(\frac{\partial}{\partial C_i} \vec{r}_j \right) = \frac{\partial}{\partial C_i} R_{i1} \cdot \vec{r}_1 + \frac{\partial}{\partial C_i} R_{i2} \cdot \vec{r}_2 + \frac{\partial}{\partial C_i} R_{i3} \cdot \vec{r}_3$$

Step 3) $[C_i, C_k] = \textcircled{1} \cdot \textcircled{2} - \textcircled{3} \cdot \textcircled{4}$ 를 정리한다.

$$\Rightarrow \left(\frac{\partial}{\partial C_i} R_{i1} \cdot \vec{r}_1 + \frac{\partial}{\partial C_i} R_{i2} \cdot \vec{r}_2 + \frac{\partial}{\partial C_i} R_{i3} \cdot \vec{r}_3 \right) \left(\frac{\partial}{\partial C_k} \vec{r}_1 + \frac{\partial}{\partial C_k} \vec{r}_2 + \frac{\partial}{\partial C_k} \vec{r}_3 \right)$$

$$- \left(\frac{\partial}{\partial C_k} R_{k1} \cdot \vec{r}_1 + \frac{\partial}{\partial C_k} R_{k2} \cdot \vec{r}_2 + \frac{\partial}{\partial C_k} R_{k3} \cdot \vec{r}_3 \right) \left(\frac{\partial}{\partial C_i} \vec{r}_1 + \frac{\partial}{\partial C_i} \vec{r}_2 + \frac{\partial}{\partial C_i} \vec{r}_3 \right) \quad (j=1, 2, 3)$$

일반적으로 (perigee가 아닐 때)

$$\vec{r}_{\text{per}} = \begin{Bmatrix} r \cos \nu \\ r \sin \nu \\ 0 \end{Bmatrix} \quad \vec{v}_{\text{per}} = \begin{Bmatrix} -na \sin \nu \\ \frac{na(1-e^2)}{\sqrt{1-e^2}} \\ 0 \end{Bmatrix} \quad \text{orbits} \quad v_3 = v_3 = 0 \quad \text{orbits}$$

$$\Rightarrow [C_i, C_k] = \sum_{j=1}^3 \left[\left(\vec{r}_1 \cdot \frac{\partial}{\partial C_i} R_{j1} \left(R_{j1} \frac{\partial}{\partial C_k} \vec{r}_1 + R_{j2} \frac{\partial}{\partial C_k} \vec{r}_2 + R_{j3} \frac{\partial}{\partial C_k} \vec{r}_3 \right) + \vec{r}_2 \frac{\partial}{\partial C_i} R_{j2} \left(R_{j1} \frac{\partial}{\partial C_k} \vec{r}_1 + R_{j2} \frac{\partial}{\partial C_k} \vec{r}_2 + R_{j3} \frac{\partial}{\partial C_k} \vec{r}_3 \right) \right) \right. \\ \left. - \left(R_{j1} \frac{\partial}{\partial C_k} \vec{r}_1 \left(\vec{r}_1 \frac{\partial}{\partial C_i} R_{j1} + \vec{r}_2 \frac{\partial}{\partial C_i} R_{j2} \right) + R_{j2} \frac{\partial}{\partial C_k} \vec{r}_2 \left(\vec{r}_1 \frac{\partial}{\partial C_i} R_{j1} + \vec{r}_2 \frac{\partial}{\partial C_i} R_{j2} + \vec{r}_3 \frac{\partial}{\partial C_i} R_{j3} \right) \right) \right]$$

2. $[C_i, C_k] = [\omega, M_0]$ 일 때는 $C_i \in \{\Omega, \omega, i\}$, $C_k \in \{a, e, M_0\}$ 인 경우이다 Case 2 이다

$$[\omega, M_0] = \sum_{j=1}^3 \left[a(1-e) \frac{\partial R_{j1}}{\partial \omega} R_{j2} \frac{\partial}{\partial M_0} \left(na \sqrt{\frac{1+e}{1-e}} \right) - \frac{na \sqrt{1-e^2}}{1-e} \frac{\partial R_{j2}}{\partial \omega} R_{j1} \frac{\partial}{\partial M_0} (a(1-e)) \right]$$

문제 1 에서 유도한 식 (at perigee) 을 이용한다 (R 에 note 57 page 참고)

$$\begin{bmatrix} \frac{\partial R_{11}}{\partial \omega} = -\cos \Omega \sin \omega - \sin \Omega \cos \omega \cos i \\ \frac{\partial R_{21}}{\partial \omega} = -\sin \Omega \sin \omega + \cos \Omega \cos \omega \cos i \\ \frac{\partial R_{31}}{\partial \omega} = \cos \omega \sin i \end{bmatrix} \quad \begin{bmatrix} \frac{\partial R_{12}}{\partial \omega} = -\cos \Omega \cos \omega + \sin \Omega \sin \omega \cos i \\ \frac{\partial R_{22}}{\partial \omega} = -\sin \Omega \cos \omega - \cos \Omega \sin \omega \cos i \\ \frac{\partial R_{32}}{\partial \omega} = -\sin \omega \sin i \end{bmatrix} \quad \begin{bmatrix} \frac{\partial R_{13}}{\partial \omega} = 0 \\ \frac{\partial R_{23}}{\partial \omega} = 0 \\ \frac{\partial R_{33}}{\partial \omega} = 0 \end{bmatrix}$$

위에서 붉은 네모에 해당하는 부분이 모두 0 이므로 $[\omega, M_0] = 0$ 이다.

따라서 $[C_i, C_k] = -[C_k, C_i]$ 관계를 생각하면 (Screw-symmetric)

$$[\omega, M_0] = -[M_0, \omega] = 0 \quad \text{이 된다.}$$